Incumbent organization and cue compensation

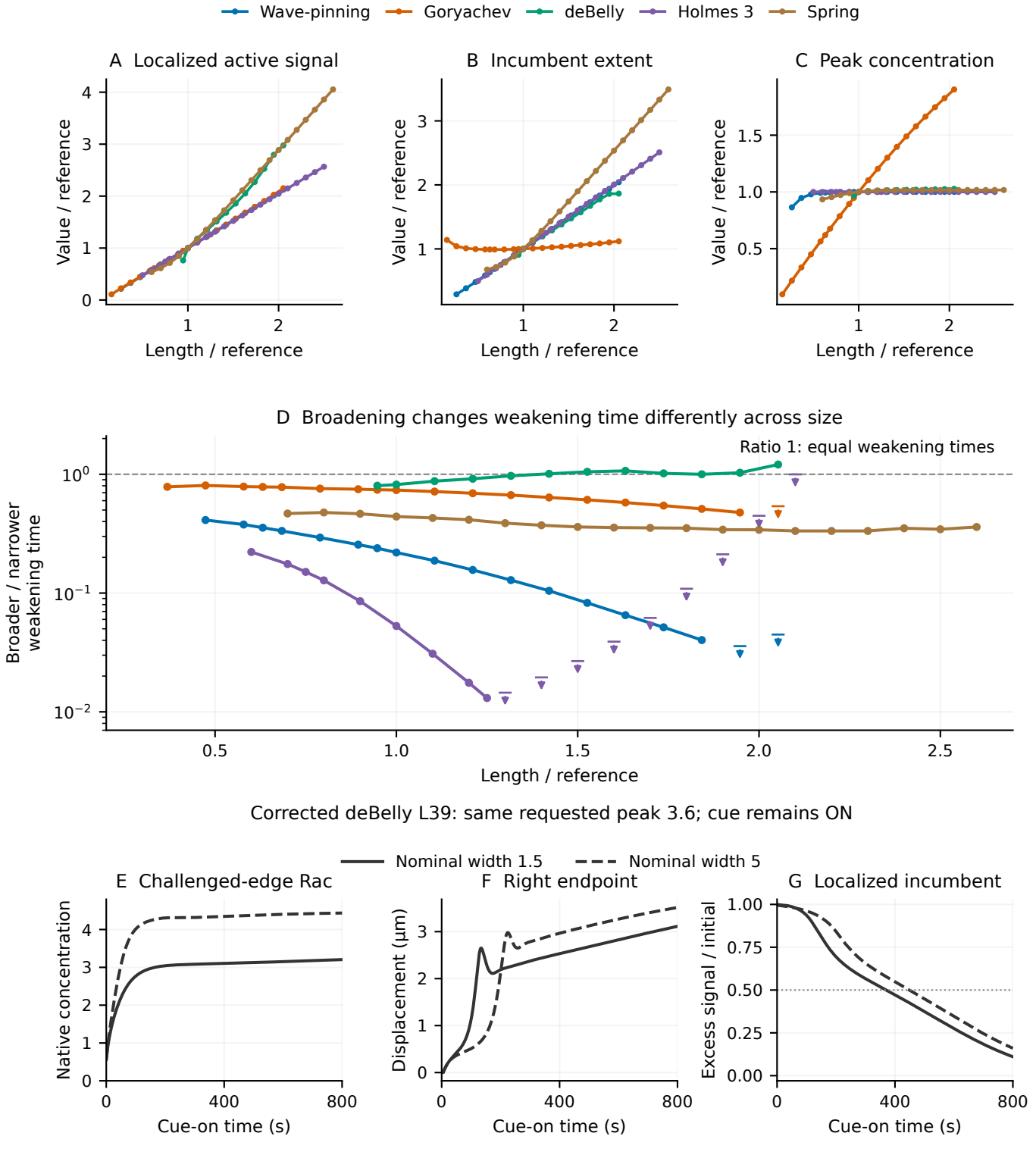


Figure 2. Incumbent organization and cue compensation

Detailed legend

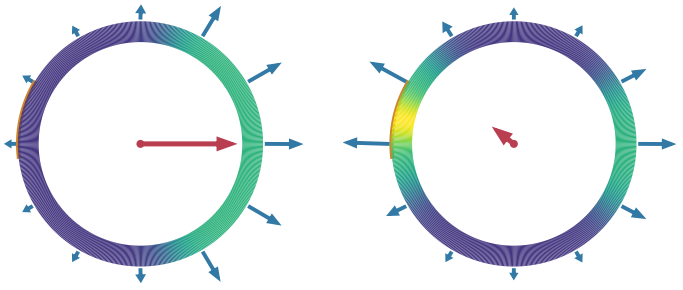
A–C, prior organization evidence. All 97 qualified basal states contain more localized active signal at greater length, through different combinations of extent and concentration. Amount is signal above the basal minimum integrated to the interpolated half-height boundary. Amount, width and peak are divided by each model's own reference value: WP/Goryachev/deBelly L19, Holmes L20, Spring resting $L_0 = 1$. Spring uses physical length; deBelly retains nominal solver coordinates. Thirteen ineligible preparations remain separately documented.

D, new broadening experiments. Within each model, the same requested native peak is applied over narrower/broader supports 1.5/5, or 0.1/0.4 in Spring. Broader inputs also increase integrated input. “Weakening time” is first sampled 50% loss of positive active signal above the current whole-profile minimum, integrated in the fixed initial organization core. Points are ratios of saved-entry bracket midpoints; thin bounds use bracket endpoints. Below one, broadening accelerates this readout; above one, it delays it. Arrows denote one-sided sampled-entry ratio bounds when one width has no entry by its finite horizon; they do not denote infinite latency or an optimized pulse threshold. Among 91 nonoverlap size pairs, 75 have two observed entries, 12 have one-sided bounds and four have neither entry. Six initial cue/core-overlap pairs remain in the source table. All 97 no-cue controls retain localized excess over their matched horizons, including 25/75% sensitivity.

E–G, a native coupled-response example. Corrected independent-endpoint deBelly L39 receives the same requested peak 3.6 over nominal widths 1.5 or 5. The common 0–800 s window remains cue-ON. Wider stimulation produces more challenged-edge Rac, an early delay followed by catch-up in endpoint displacement, and later 50% incumbent-excess loss: (370,372.5] versus (447.5,450] s. These readouts are distinct. Both widths return after a tested 500 s pulse; both 800/2000 s pulses pass retained scalar-strength and native field-tail criteria. Earlier regional weakening therefore does not establish a different retention fate at 500 s. Scalar gates alone do not prove whole-field replacement.

The comprehensive supplement retains full strength/duration outcomes, complete histories, background/core sensitivity and native physical cue mappings. Current-profile floors describe localized signal, not a molecular pool. The example concerns native coupled activity and mechanics, not isolated causal mediation or a calibrated beam-width prediction. Original discretizations and finite observation windows remain explicit in the detailed methods.

A Local activity adds to a global vector

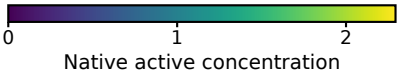


Holmes · C=30
d_0=14 · T=300

$M_0 = C\langle u \rangle$

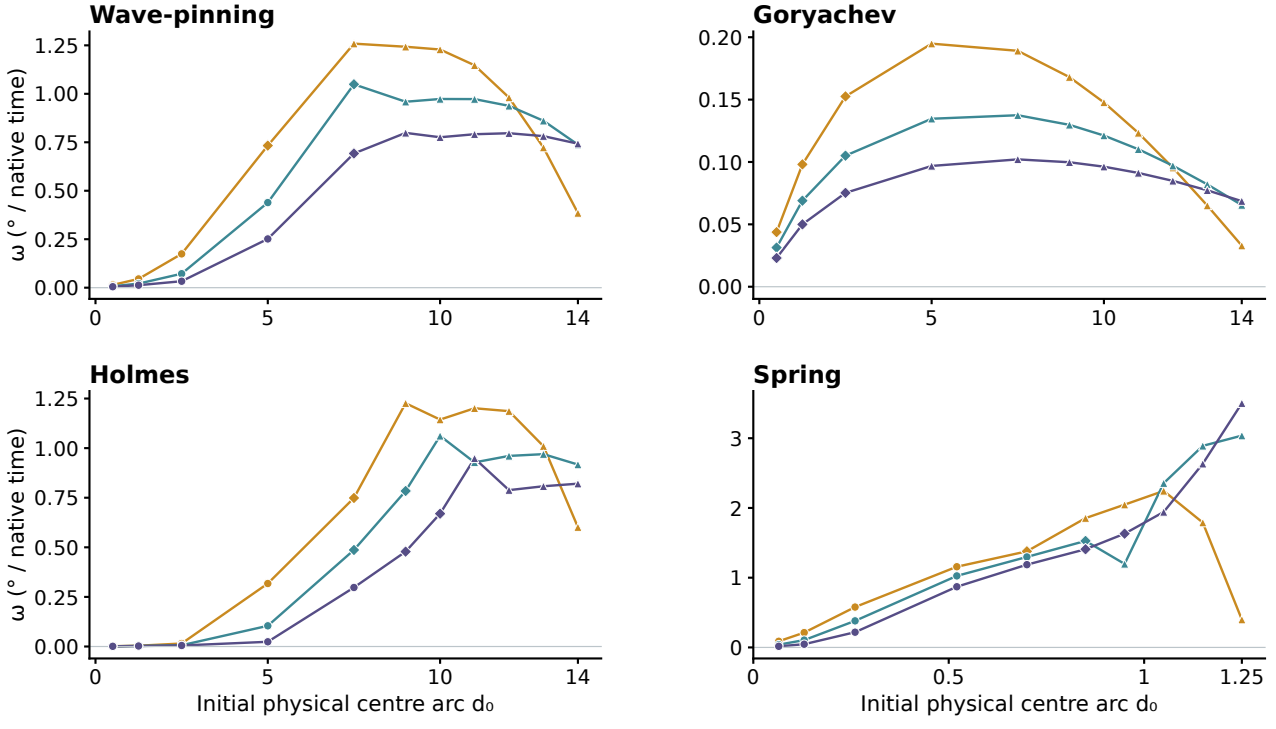
$M_1 = C\langle u e^{2\pi i q} \rangle$

$P = M_1/M_1(0), \quad R = |P|$
Blue: sector contributions
Red: resultant P



B Cue-on angular rate at t=10





C Wave-pinning: the same external physical cue

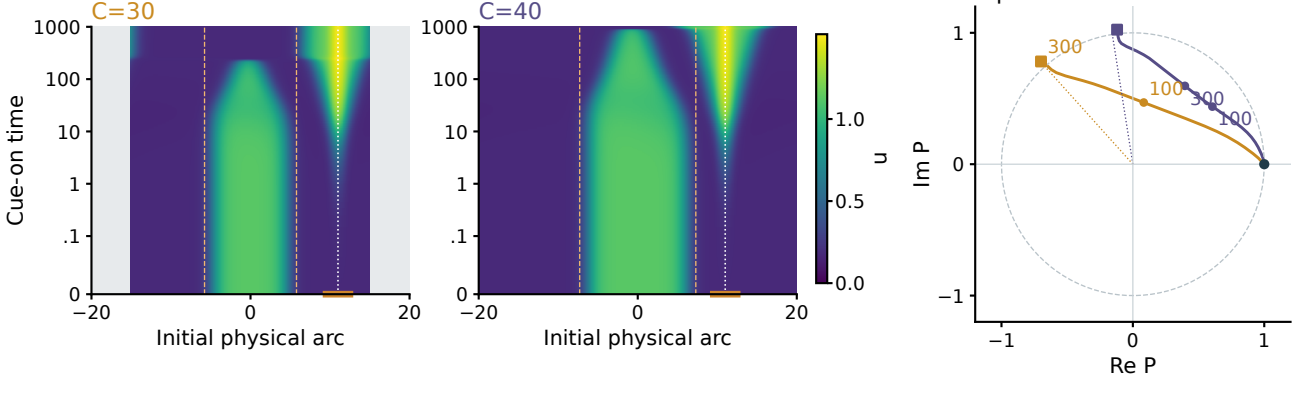


Figure 3. Stationary cue placement and activity-vector reorientation

Detailed legend

Figure 3. Stationary cue placement and native size shape global activity-vector reorientation. Four native periodic models are challenged from exact qualified polarized basals, using chemical circumferences 30/35/40 and Spring resting circumferences 2.5/2.75/3. Stationary localized cues preserve native kinetics, input channels, coefficients and physical supports. Initial physical centre arc d_0 is prescribed from the preparation axis; all internal and overlapping placements remain. The completed experiment contains 645 protocols: 175 exact archive-pair reuses and 470 fresh branches.

A, Measured vector construction. Local active concentration is a real scalar. Weighting each physical active amount by its outward unit normal gives two Cartesian components, represented together by the complex first moment $M_1 = C\langle ue^{2\pi iq}\rangle$. Normalization $P = M_1/M_1(0)$ rotates the reference axes once and sets $P(0) = 1$; $R = |P|$. Actual Holmes C=30 profiles before and after a 300-unit cue at d_0=14 show how directional contributions partly cancel. Blue arrows are exact normalized native-cell sums in twelve 30° sectors, translated for display; red is their sum, with one shared vector scale. Active amount rises to 1.029 of baseline while R falls to 0.259. Colours share a zero-based native concentration scale; orange arcs mark cue supports. This is a cue-on state, without a stationary-fate or site-ancestry inference. Its later cue-free field tail fails the finite gate.

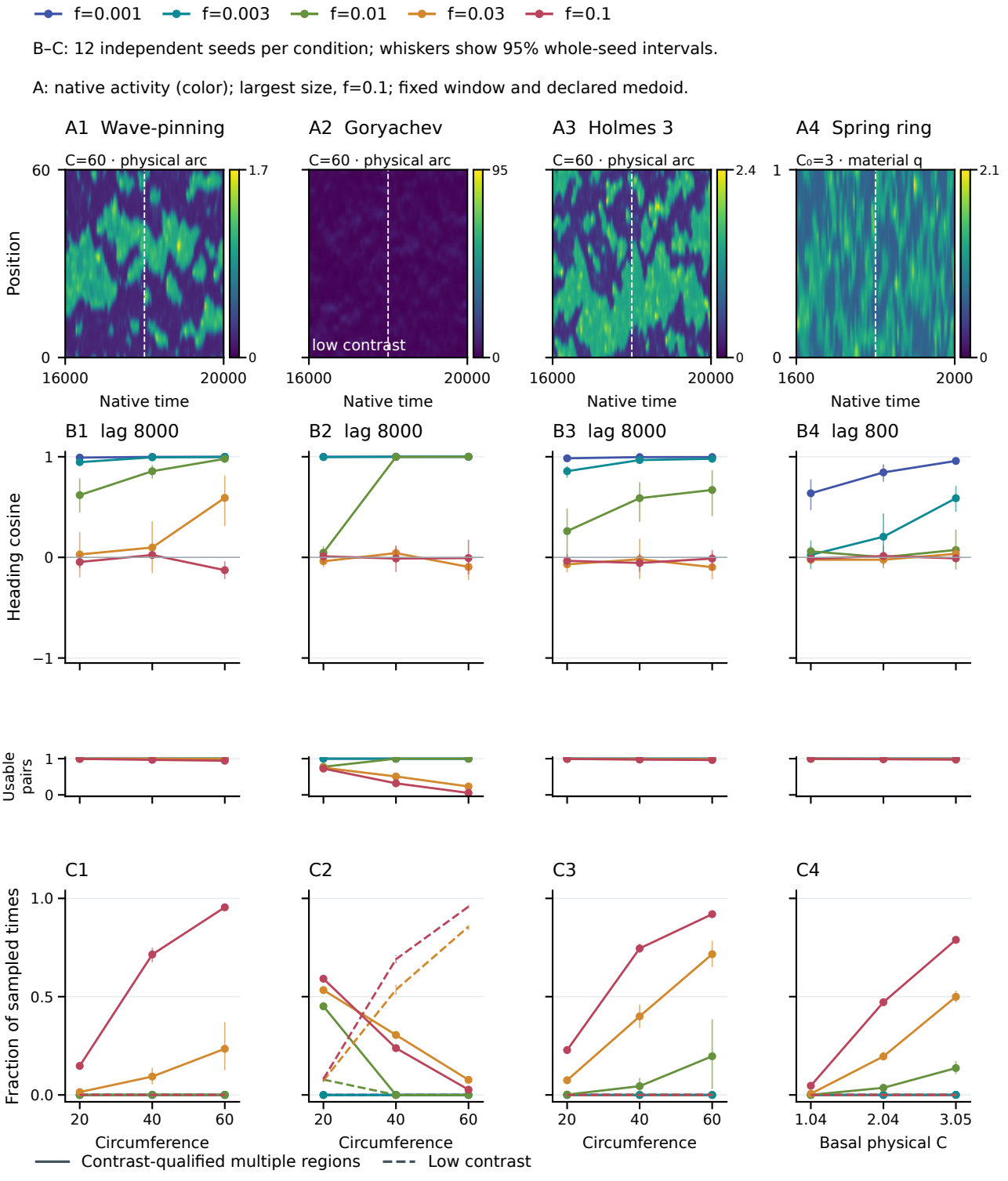
B, Complete placement comparison. Early instantaneous native-RHS angular rate at native time 10 is shown for all 129 placements using their 1000-unit cue branches. Colour identifies native size, with chemical C/Spring C_0 pairs in the shared key. Circles, diamonds and triangles classify complete initial footprints inside, straddling or outside the interpolated 10%-contrast region. Lines connect tested points without fitting a distance law. All 129 displayed instants retain full-state/snapshot support and R≥0.1. Nonmonotone distance profiles and changing size ranks demonstrate why global angular rate requires spatial context. Full temporal and duration responses remain in the supplement.

C, Native field history and resultant. Wave-pinning C=30/40 basals receive the same fully external cue at d_0=11 for 1000 units. Every saved field uses common physical-arc and native-amplitude scales; grey padding is outside the shorter domain. Dashed lines mark initial 10%-contrast edges; white lines/orange bars mark cue centres/supports. Time is linear through 0.1 then logarithmic. Equal-scale complex-P paths retain direction and magnitude; the circle is R=1, squares mark withdrawal and numbers give actual native times. Dotted rays mark actual targets. Targets differ (132°/99°), so architecture and angular geometry change together.

Rates describe an activity resultant rather than peak or cell velocity. Initial projection and moment magnitude matter; cross-model clocks are uncalibrated. Complete duration grids, definitions, weak-vector masking, all 645 outcomes, native profiles and finite 13,000-unit cue-free tail qualifications are reported in the comprehensive supplement; no causal geometry partition or asymptotic-stability claim follows.

Figure 4 · Input regime shapes size effects on activity memory

Native local organization and second-half summaries across four ring models



Direction requires both endpoint R≥0.1; the thin strip retains all-pair coverage.

Contrast is qualified at ≥10% of basal. Full profiles, matched mean controls and windows: supplement.

Native clocks/laws differ. Whole positive input scales with f. Activity heading is distinct from migration.

First half discarded without assuming stationarity. Spring sampled run-duration inference remains excluded.

Figure 4. Input regime, circumference and activity-heading memory

Detailed legend

Figure 4. Input regime shapes size effects on activity-heading memory and local organization. Columns show Wave-pinning, Goryachev, Holmes 3 and Spring ring. Each displayed model–size–input condition has twelve independent noisy realizations. Increasing f scales the whole positive activation signal, including its mean and fluctuation amplitude. Native clocks and forcing references differ across models. The same normalized physical-activity vector P as Figure 3 is used; R=|P|. Activity heading is distinct from cell migration.

A1–A4: native activity during a fixed short second-half window at the largest original size and f=0.1. Histories are declared morphology-descriptor medoids, with zero-based model-native concentration scales. White dashed vertical lines mark fixed snapshots at chemical time 18000 or Spring time 1800; matching actual profiles, basals and deterministic ensemble-mean controls are in the supplementary native-profile sheet. The displayed chemical window is 16000–20000 and Spring window 1600–2000. Spring position is material coordinate q; its evolving physical circumference remains in all activity moments.

B1–B4: pooled cosine of activity-heading differences at native lag 8000 for chemicals or 800 for Spring, conditional on both endpoint magnitudes R≥0.1. Narrow strips show eligible pairs divided by every available lag pair, so loss of usable direction remains visible. Whiskers are pointwise 95% percentile intervals from 2000 whole-seed resamples, paired across readouts and windows. Time samples are not treated as independent replicates. Full temporal curves with coverage and explicit sparse-support rules remain in the supplement and methods.

C1–C4: solid curves show the fraction of all scalar times with multiple periodic connected arcs above current minimum plus half current contrast (maximum minus minimum), additionally requiring contrast≥10% of the same case’s basal contrast. Dashed curves show the fraction below that contrast floor. These descriptors distinguish contrast-rich multiple-region organization from low-contrast states without assigning functional competition or exclusive phases. Summaries discard the first half, using chemical times 16000–32000 or Spring 1600–3200; window checks retain observed sensitivity rather than assuming equilibrium.

The broad-grid experiment comprises 816 noisy realizations, 108 matched-mean controls and 12 no-input controls: 936 scientific cases. Five strengths have twelve seeds; four outer strengths retain two exploratory seeds. Comprehensive support retains every model, size and strength, native fields, activity amount/coherence, controls, spatial widths and temporal/window evidence. Methods detail the Goryachev mean-control qualifications and the unchanged Spring sampled-duration failure; the main readouts use the separately accepted scope. A separate Goryachev C60 refinement adds 24 noisy realizations and two matched-mean controls at f=0.015/0.02 (supplement).